SIMPLIFY WITH THE GREATEST COMMON FACTOR

Name: ..... **Date:** .....

List the factors of both numbers, find the greatest factor they share (the GCF), then divide both by it.

1. $\frac{15}{20}$ — Factors of 15: ____ Factors of 20: ____ GCF ____ = ____
2. $\frac{12}{30}$ — Factors of 12: ____ Factors of 30: ____ GCF ____ = ____
3. $\frac{8}{14}$ — Factors of 8: ____ Factors of 14: ____ GCF ____ = ____
4. $\frac{4}{6}$ — Factors of 4: ____ Factors of 6: ____ GCF ____ = ____
5. $\frac{4}{16}$ — Factors of 4: ____ Factors of 16: ____ GCF ____ = ____
6. $\frac{2}{8}$ — Factors of 2: ____ Factors of 8: ____ GCF ____ = ____
7. $\frac{2}{14}$ — Factors of 2: ____ Factors of 14: ____ GCF ____ = ____
8. $\frac{5}{15}$ — Factors of 5: ____ Factors of 15: ____ GCF ____ = ____
9. $\frac{2}{10}$ — Factors of 2: ____ Factors of 10: ____ GCF ____ = ____
10. $\frac{4}{10}$ — Factors of 4: ____ Factors of 10: ____ GCF ____ = ____

Worked example: $\frac{12}{18}$: factors of 12 are 1, 2, 3, 4, 6, 12; of 18 are 1, 2, 3, 6, 9, 18. GCF = 6, so $\frac{12}{18} = \frac{2}{3}$.

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SIMPLIFY WITH THE GREATEST COMMON FACTOR ANSWER KEY

Name: ..... **Date:** .....

List the factors of both numbers, find the greatest factor they share (the GCF), then divide both by it.

1. $\frac{15}{20}$
   Factors of 15: 1, 3, 5, 15
   Factors of 20: 1, 2, 4, 5, 10, 20
   GCF = 5 → $\frac{3}{4}$

2. $\frac{12}{30}$
   Factors of 12: 1, 2, 3, 4, 6, 12
   Factors of 30: 1, 2, 3, 5, 6, 10, 15, 30
   GCF = 6 → $\frac{2}{5}$

3. $\frac{8}{14}$
   Factors of 8: 1, 2, 4, 8
   Factors of 14: 1, 2, 7, 14
   GCF = 2 → $\frac{4}{7}$

4. $\frac{4}{6}$
   Factors of 4: 1, 2, 4
   Factors of 6: 1, 2, 3, 6
   GCF = 2 → $\frac{2}{3}$

5. $\frac{4}{16}$
   Factors of 4: 1, 2, 4
   Factors of 16: 1, 2, 4, 8, 16
   GCF = 4 → $\frac{1}{4}$

6. $\frac{2}{8}$
   Factors of 2: 1, 2
   Factors of 8: 1, 2, 4, 8
   GCF = 2 → $\frac{1}{4}$

7. $\frac{2}{14}$
   Factors of 2: 1, 2
   Factors of 14: 1, 2, 7, 14
   GCF = 2 → $\frac{1}{7}$

8. $\frac{5}{15}$
   Factors of 5: 1, 5
   Factors of 15: 1, 3, 5, 15
   GCF = 5 → $\frac{1}{3}$

9. $\frac{2}{10}$
   Factors of 2: 1, 2
   Factors of 10: 1, 2, 5, 10
   GCF = 2 → $\frac{1}{5}$

10. $\frac{4}{10}$
    Factors of 4: 1, 2, 4
    Factors of 10: 1, 2, 5, 10
    GCF = 2 → $\frac{2}{5}$

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